

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

M2 Practical 9

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Aim: Hypothesis Testing and A/B Testing

1. Load the supermarket_sales.csv dataset.:

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("supermarket_sales.csv")
```

2. Display the first five records:

```
print("First five records:")
print(df.head())
```

```
First five records:
Invoice ID Branch City Customer type Gender \
0 750-67-8428 A Yangon Member Female
1 226-31-3081 C Naypyitaw Normal Female
2 631-41-3108 A Yangon Normal Male
3 123-19-1176 A Yangon Member Male
4 373-73-7910 A Yangon Normal Male

Product line Unit price Quantity Tax 5% Total Date \
0 Health and beauty 74.69 7 26.1415 548.9715 1/5/2019
1 Electronic accessories 15.28 5 3.8200 80.2200 3/8/2019
2 Home and lifestyle 46.33 7 16.2155 340.5255 3/3/2019
3 Health and beauty 58.22 8 23.2880 489.0480 1/27/2019
4 Sports and travel 86.31 7 30.2085 634.3785 2/8/2019

Time Payment cogs gross margin percentage gross income Rating
0 13:08 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4
3 20:33 Ewallet 465.76 4.761905 23.2880 8.4
4 10:37 Ewallet 604.17 4.761905 30.2085 5.3
```

3. Select the revenue column:

```
revenue = df["Total"]
```

5. Calculate the mean revenue:

```
mean_revenue = revenue.mean()
```

```
print("\nMean Revenue:", mean_revenue)
```

Mean Revenue: 322.966749

6. Assume that the expected average revenue is ₹1,500:

```
expected_mean = 1500
```

7. Write the null hypothesis: $H_0: \mu = 1500$

8. Write the alternative hypothesis: $H_1: \mu \neq 1500$

```
print("\nRevenue:")
print(revenue)
```

```
Revenue:
0      548.9715
1       80.2200
2      340.5255
3      489.0480
4      634.3785
...
995     42.3675
996    1022.4900
997     33.4320
998     69.1110
999     649.2990
Name: Total, Length: 1000, dtype: float64
```

4. Check for missing values:

```
print("\nMissing values in revenue:")
print(revenue.isnull().sum())
```

```
# Remove missing values if present
revenue = revenue.dropna()
```

Missing values in revenue:

0

```
# 7. Alternative Hypothesis
# H1: pA != pB

# 8. Significance level
alpha = 0.05

✓ 0.0s
```

9. Set the significance level as 5%:

```
alpha = 0.05
```

10. Perform a one-sample t-test:

```
t_stat, p_value = stats.ttest_1samp(
    revenue,
    expected_mean
)
```

11. Calculate the p-value:

```
print("\nT-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -151.37567896776733
P-value: 0.0
```

12 & 13. Compare p-value with 0.05:

```
if p_value < alpha:
    print("Reject the Null Hypothesis")
else:
    print("Do not Reject the Null Hypothesis")
```

```
Reject the Null Hypothesis
```

Conclusion: The test shows that the average revenue is significantly different from ₹1,500. Therefore, the business should not assume that its average revenue is ₹1,500 based on the given data.

B. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

```
import pandas as pd
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt
```

1. Load the dataset

```
data = pd.read_csv("ab_testing.csv")
```

```
# Convert Conversion column into numeric values
# If values are 0 and 1 stored as text, this will convert them
data["Conversion"] = pd.to_numeric(data["Conversion"],
errors="coerce")
```

2. Display the data

```
print("A/B Testing Dataset:")
print(data)
```

A/B Testing Dataset:			
	User	Group	Conversion
0	14292	B	NaN
1	11682	A	NaN
2	19825	A	NaN
3	16080	B	NaN
4	18851	A	NaN
...	...	...	...
4995	16360	B	NaN
4996	18084	B	NaN
4997	12063	A	NaN
4998	18647	B	NaN
4999	16686	A	NaN

```
# Check for missing values
data = data.dropna(subset=["Conversion"])
```

3. Calculate conversion rate for Group A

```
group_A = data[data["Group"] == "A"]
```

```
users_A = len(group_A)
conversions_A = group_A["Conversion"].sum()
```

```
rate_A = conversions_A / users_A
```

```
print("\nGroup A Conversion Rate:", rate_A)
print("Group A Conversion Rate:", rate_A * 100, "%")
```

```
Group A Conversion Rate: nan
Group A Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_...
rate_A = conversions_A / users_A
```

4. Calculate conversion rate for Group B

```
group_B = data[data["Group"] == "B"]
```

```
users_B = len(group_B)
conversions_B = group_B["Conversion"].sum()
```

```
rate_B = conversions_B / users_B
```

```
print("\nGroup B Conversion Rate:", rate_B)
print("Group B Conversion Rate:", rate_B * 100, "%")
```

```
Group B Conversion Rate: nan
Group B Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_...
rate_B = conversions_B / users_B
```

5. Compare conversion rates

```
print("\nComparison:")
print("Group A:", rate_A * 100, "%")
print("Group B:", rate_B * 100, "%")
```

```
if rate_A > rate_B:
    print("Group A has a higher conversion rate.")
elif rate_B > rate_A:
    print("Group B has a higher conversion rate.")
else:
    print("Both groups have the same conversion rate.")
```

```
Comparison:
Group A: nan %
Group B: nan %
Both groups have the same conversion rate.
```

6. Null Hypothesis

```
print("\nH0: pA = pB")
```

```
H0: pA = pB
```

7. Alternative Hypothesis

```
print("H1: pA != pB")
```

```
H1: pA != pB
```

8. Significance level

```
alpha = 0.05
print("Significance Level:", alpha)
```

```
Significance Level: 0.05
```

9. Perform two-proportion test

```
conversions = [conversions_A, conversions_B]
users = [users_A, users_B]
```

```
z_stat, p_value = proportions_ztest(conversions, users)
```

```
c:\Users\itlab\AppData\Local\Programs\Python\Py
prop = count * 1. / nobs
c:\Users\itlab\AppData\Local\Programs\Python\Py
p_pooled = np.sum(count) * 1. / np.sum(nobs)
c:\Users\itlab\AppData\Local\Programs\Python\Py
nobs_fact = np.sum(1. / nobs)
```

10. Calculate and display p-value

```
print("\nZ-statistic:", z_stat)
print("P-value:", p_value)
```

```
Z-statistic: nan
P-value: nan
```

11. Compare p-value with 0.05

```
print("\nComparison of p-value with 0.05:")
```

```
if p_value < alpha:
    print("p-value < 0.05")
else:
    print("p-value >= 0.05")
```

Conclusion:

The conversion rate of Group A and Group B was calculated and compared using a two-proportion z-test at a 5% significance level. Based on the p-value, the null hypothesis is not rejected if the p-value is greater than 0.05. Therefore, there is not enough statistical evidence to conclude that the conversion rates of Group A and Group B are significantly different.

```
Comparison of p-value with 0.05:
p-value >= 0.05
```

12. Decision

```
if p_value < alpha:
    print("Decision: Reject the Null Hypothesis (H0)")
else:
    print("Decision: Do not reject the Null Hypothesis (H0)")
```

```
Decision: Do not reject the Null Hypothesis (H0)
```

13. Create bar chart

```
groups = ["Group A", "Group B"]
conversion_rates = [rate_A * 100, rate_B * 100]
```

```
plt.bar(groups, conversion_rates)
```

```
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing - Conversion Rate Comparison")
# Display values on bars
for i, rate in enumerate(conversion_rates):
    plt.text(i, rate, f"rate:.2f%", ha="center", va="bottom")
```

```
plt.show()
```

